

LAB EXPERIMENT RECORD

Set 3: Employee Performance Evaluation

Department	AIDS
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	DIVYASREE.M
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Aim

To analyse employee performance data using a line chart, bar chart and scatter plot, and to build an interactive dashboard combining the line and bar charts using R.

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

Procedure

1. Create vectors (or a data frame) for employee ID, department, years of service and performance score.
2. Use `plot()` with `type = "o"` and `legend()` to draw the labelled performance line chart.
3. Count employees per department with `table()` and draw the bar chart using `barplot()`.
4. Use `plot()` to draw years of service against performance and `cor()` to measure the correlation.
5. Load the shiny package, build the UI with department and minimum-score filters, and write server logic using `reactive()` and `renderPlot()`.
6. Run the app with `shinyApp()` and use the filters to explore both charts and the records table interactively.

Key R Functions Used

Function	Purpose
<code>plot()</code> / <code>legend()</code>	Create line and scatter plots with a legend.
<code>table()</code> / <code>barplot()</code>	Count and plot employees per department.
<code>cor()</code>	Computes the correlation coefficient.
<code>selectInput()</code> / <code>numericInput()</code>	Provide department and score filters.
<code>reactive()</code> / <code>renderPlot()</code> / <code>renderTable()</code>	Update charts and table dynamically.
<code>shinyApp()</code>	Runs the interactive dashboard.

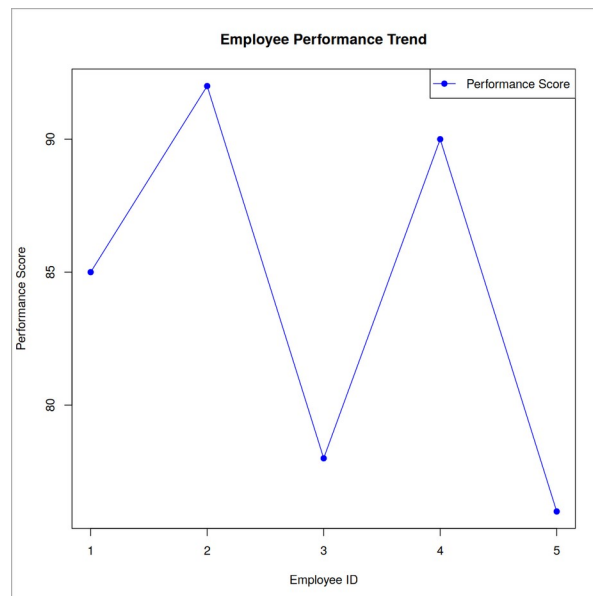
PROGRAM 1: Line Chart of Employee Performance

Create a line chart to visualize the performance trend of employees. Include a legend and labels.

```
# Dataset
emp_id      <- 1:5
performance <- c(85, 92, 78, 90, 76)

# Line Chart
plot(emp_id, performance,
     type = "o",
     col = "blue",
     pch = 19,
     xlab = "Employee ID",
     ylab = "Performance Score",
     main = "Employee Performance Trend")

# Legend
legend("topright",
      legend = "Performance Score",
      col = "blue", pch = 19, lty = 1)
```



Output 1: Employee Performance Trend (Line Chart)

Observation: Employee 2 has the highest score (92) and Employee 5 the lowest (76). Scores alternate between high and low values across employees, ranging from 76 to 92 with an average of 84.2.

Note: The dataset has no time column, so performance is plotted across Employee IDs.

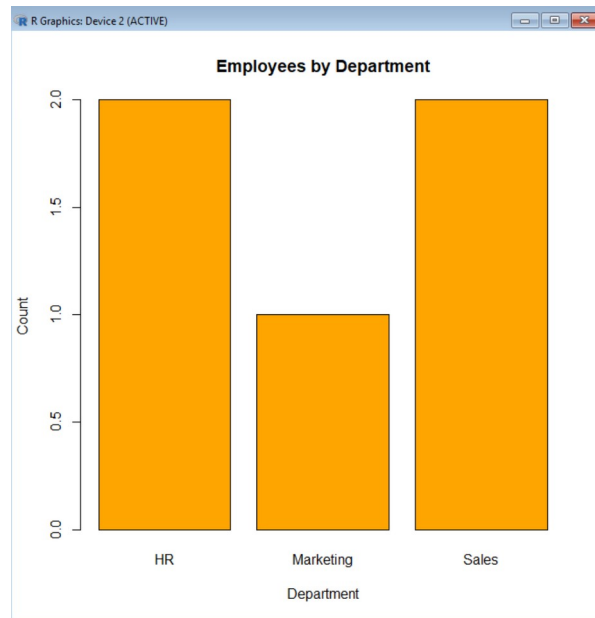
PROGRAM 2: Bar Chart of Employees by Department

Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.

```
# Dataset
department <- c("Sales", "HR", "Marketing", "Sales", "HR")

# Count employees in each department
dept_count <- table(department)

# Bar Chart
barplot(dept_count,
        col = "orange",
        main = "Employees by Department",
        xlab = "Department",
        ylab = "Count")
```



Output 2: Employees by Department (Bar Chart)

Observation: HR and Sales have two employees each, while Marketing has one. Average performance is highest in Sales (87.5), followed by HR (84) and Marketing (78).

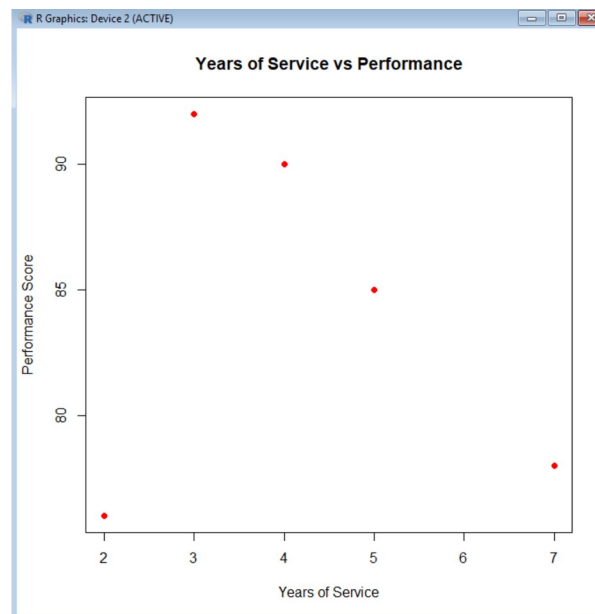
PROGRAM 3: Scatter Plot of Years of Service vs Performance

Build a scatter plot to analyze the correlation between years of service and performance scores. Explain any insights.

```
# Dataset
years      <- c(5, 3, 7, 4, 2)
performance <- c(85, 92, 78, 90, 76)

# Scatter Plot
plot(years, performance,
     pch = 19,
     col = "red",
     xlab = "Years of Service",
     ylab = "Performance Score",
     main = "Years of Service vs Performance")

# Correlation
cor(years, performance)    # -0.17
```



Output 3: Years of Service vs Performance (Scatter Plot)

Insight: The correlation is -0.17, which means there is no strong linear relationship between years of service and performance. The best scores (92 and 90) come from employees with 3–4 years of service, while both the newest employee (2 years, 76) and the most experienced employee (7 years, 78) score lowest. Experience alone does not predict performance in this dataset.

PROGRAM 4: Interactive Employee Performance Dashboard

Develop a dashboard with the line chart and bar chart for interactive exploration.

```
library(shiny)

# Dataset
emp <- data.frame(
  EmployeeID = 1:5,
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  Years      = c(5, 3, 7, 4, 2),
  Performance = c(85, 92, 78, 90, 76)
)
dept_cols <- c(HR = "red", Marketing = "darkgreen", Sales = "blue")
```

```

ui <- fluidPage(
  titlePanel("Employee Performance Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      selectInput("dept", "Filter by Department:",
        choices = names(dept_cols),
        selected = names(dept_cols),
        multiple = TRUE),
      numericInput("score", "Minimum Performance Score:",
        value = 70, min = 70, max = 95, step = 1)
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("linePlot", height = "350px")),
        column(6, plotOutput("barPlot", height = "350px"))
      ),
      h4("Filtered Employee Records"),
      tableOutput("table")
    )
  )
)

server <- function(input, output) {

  filtered <- reactive({
    emp[emp$Department %in% input$dept & emp$Performance >= input$score, ]
  })

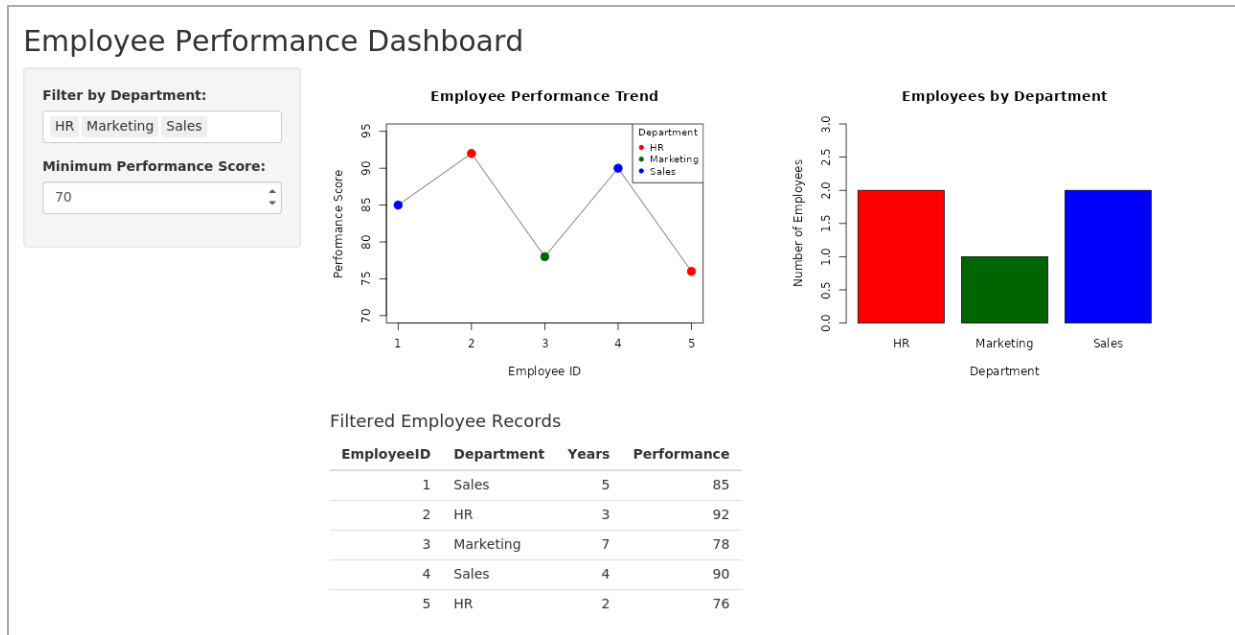
  output$linePlot <- renderPlot({
    d <- filtered()
    req(nrow(d) > 0)
    plot(d$EmployeeID, d$Performance, type = "o", pch = 19, col = "gray40",
      xlim = c(1, 5), ylim = c(70, 95), xaxt = "n",
      xlab = "Employee ID", ylab = "Performance Score",
      main = "Employee Performance Trend")
    points(d$EmployeeID, d$Performance, pch = 19, cex = 1.6,
      col = dept_cols[d$Department])
    axis(1, at = 1:5)
    legend("topright", legend = names(dept_cols), col = dept_cols,
      pch = 19, title = "Department", cex = 0.85)
  }, width = 420, height = 340)

  output$barPlot <- renderPlot({
    d <- filtered()
    req(nrow(d) > 0)
    counts <- table(factor(d$Department, levels = names(dept_cols)))
    barplot(counts, col = dept_cols, ylim = c(0, 3),
      xlab = "Department", ylab = "Number of Employees",
      main = "Employees by Department")
  }, width = 420, height = 340)

  output$table <- renderTable(filtered(), digits = 0)
}

shinyApp(ui = ui, server = server)

```



Output 4: Interactive Employee Performance Dashboard (R Shiny)

Observation: The dashboard combines the performance line chart (points coloured by department, with a legend) and the department bar chart, along with a table of the filtered records. Choosing departments or raising the minimum score updates all three views together — for example, setting the minimum score to 85 leaves only Employees 1, 2 and 4.

Note: The question specifies a Tableau dashboard. Since this record is implemented in R, the equivalent interactive dashboard was built with R Shiny. In Tableau the same result is obtained by importing the dataset, creating a line chart sheet (Employee ID vs Performance Score) and a bar chart sheet (Department vs COUNT of Employee ID), placing both on a dashboard, and adding a Department filter applied to both sheets.

Result

The employee performance data was successfully visualized in R using a line chart, bar chart and scatter plot, and an interactive Shiny dashboard combining the line and bar charts was created and executed.